

Gowtham Chakri Mallepaka

Software Engineer / Machine Learning Engineer

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SUMMARY

Experienced Software Engineer with over 5+ years of expertise in Machine Learning Engineer, Data Science, Generative AI, Computer Vision, and predictive modeling. Proficient in PyTorch, advanced data management techniques, and developing scalable machine learning algorithms. Skilled in deploying and optimizing machine learning models for real-world applications, ensuring operational efficiency and driving impactful outcomes. Demonstrates a strong ability to troubleshoot complex technical challenges and deliver innovative solutions across global projects. Skilled in applying quantitative and statistical analysis to enhance industrial processes, drive efficiency improvements, and deliver actionable insights that support informed decision-making in fast-moving environments.

PROFESSIONAL EXPERIENCE

ML Software Engineer | Meta | USA

Oct 2022 – Present

- Developed and implemented a web-based Data Asset Management System using React and Vue.js, achieving a 40% reduction in development time and enhancing application performance.
- Provided remote technical support for data asset management systems, maintaining 90% issue resolution within SLA and ensuring seamless operations across platforms.
- Integrated Continuous Integration/Deployment (CI/CD) pipelines into Amazon Pipelines and AWS SaaS infrastructure, reducing deployment errors by 30% and significantly improving operational efficiency.
- Led the development of machine learning models using Generative AI, employing BERT and GPT-based architectures for NLP tasks and integrating a Retrieval-Augmented Generation (RAG) Chatbot to enhance clarity and user experience by 50%.
- Engineered real-time fraud detection pipelines using Spark and AWS Kinesis, incorporating anomaly detection models that reduced false positives by 25% and improved transaction security for over 1 million daily users.
- Optimized feature engineering workflows using automated pipelines with MLflow and Airflow, accelerating model retraining cycles by 35% and boosting model accuracy in credit risk scoring applications.
- Collaborated with cross-functional teams (DevOps, data engineering, and cybersecurity) to design secure, scalable ML model APIs using FastAPI and Docker, decreasing inference latency by 40% while ensuring compliance with PNC's data governance policies.
- Architected and rolled out scalable backend and frontend systems using Qt with C++, reducing system crashes by 60% and improving performance by 20% on ARM437X microprocessors for energy sector applications.
- Engineered secure APIs with Django REST Framework, incorporating serialization and token-based authentication to ensure data integrity and security.
- Enhanced GPU performance by 20% through advanced communication protocols such as SPI and MQTT, driving efficiency in embedded systems within microservices-based architectures.
- Streamlined development processes by integrating version control, CI/CD pipelines, and automation tools, leading to a 15% increase in productivity and a 10% reduction in production release errors.
- Performed regular code reviews, providing constructive feedback to enforce coding standards, resulting in a 15% improvement in overall code quality.
- Developed and optimized multithreaded applications for real-time system performance, minimizing latency issues and enhancing system responsiveness in industrial control environments.

ML Software Engineer | Wipro Ltd | India

Aug 2020 – Sep 2022

- Worked in an Agile environment collaborating with other teams to enhance workflow predictability and cut down the delivery time variability by 25% for machine learning projects.
- Performed data operations which included collecting, cleaning, mining, hypothesis testing, building models, validating and visualizing the data and has led to improving the quality of insights generated by 30%.
- Utilized Python libraries including Pandas, NumPy, Scikit-learn, TensorFlow, Spark, Matplotlib, and SciPy in developing efficient machine learning models that has improved operational efficiency by 20%.
- Structured and developed predictive models using some models and algorithms including Linear Regression, Logistic Regression, Decision Trees, Random Forest, AdaBoost, XGBoost, Support Vector Machines (SVM), K-Nearest Neighbors (KNN) thus enhancing insurance predictions by 35%.
- Implemented feature engineering techniques such as one-hot encoding, normalization, and dimensionality reduction (PCA, t-SNE), leading to a 15% boost in model accuracy and reduced training time.
- Designed and maintained automated ML pipelines using Airflow and Spark for continuous training and deployment, reducing manual intervention and accelerating model delivery cycles by 40%.

EDUCATION

Kent State University, OH, USA

Master of Science in Business Analytics

Aug 2023

Osmania University, Hyderabad, India

Master of Business Administration

Oct 2021

SKILLS

Programming Languages: Python, R, Java, C++, Scala, SQL, MATLAB, Bash/Shell Scripting

Machine Learning: Supervised, Unsupervised, Deep Learning, Feature Engineering, Model Training, Model Validation, Model Optimization, Hyperparameter Tuning, Regression & Classification - Linear, Polynomial, Logistic Regression, Decision Tree, Random Forest, Clustering (K-Means, Hierarchical), Reinforcement Learning (UCB, Thompson Sampling), Association Rule Learning (Apriori, Eclat), Sentiment Analysis, Topic Modelling
AI/NLP: Large Language Models (LLMs), NLP, Text Analytics, Transformer Models, BERT, GPT, Hugging Face, SpaCy

Computer Vision: OpenCV, TensorFlow, PyTorch, CNNs, Image Segmentation, Object Detection

Big Data: Apache Spark, Hadoop, Distributed File Systems, Data Ingestion, ETL, Data Pipelines

Cloud Platforms: AWS (Sage Maker, EC2, S3, Lambda), GCP, Azure

DevOps/CI-CD: Docker, Kubernetes, Jenkins, Terraform, MLflow, TensorBoard, Git/GitHub

Deployment & Monitoring: Model Deployment (MLflow, Windows ML, Kubeflow, Triton), Model Drift Detection